

Data Cleaning & Visualization Report

Generated on: 2026-04-25 12:47:49

Dataset: Titanic (Seaborn)

1. Data Inspection & Initial State

Original Dataset Shape: 891 rows x 15 columns

Duplicate Rows Found: 107

Data Preview (First 5 Rows):

survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	3	male	22.0	1	0	7.25	S	Third	man	True	nan	Southampton	no	False
1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
1	3	female	26.0	0	0	7.925	S	Third	woman	False	nan	Southampton	yes	True
1	1	female	35.0	1	0	53.1	S	First	woman	False	C	Southampton	yes	False
0	3	male	35.0	0	0	8.05	S	Third	man	True	nan	Southampton	no	True

Missing Values Summary:

Column	Null Count	Percentage
age	177	19.87%
embarked	2	0.22%
deck	688	77.22%
embark_town	2	0.22%

2. Descriptive Statistics

Feature	count	unique	top	mean	std	min	max
survived	891.0	-	-	0.38383 838383 83	0.48659 245426 48	0.0	1.0
pclass	891.0	-	-	2.30864 197530 86	0.83607 124097 70	1.0	3.0
sex	891.0	2	male	-	-	-	-
age	714.0	-	-	29.6991 176470 58	14.5264 973323 34	0.42	80.0
sibsp	891.0	-	-	0.52300 785634 11	1.10274 343229 34	0.0	8.0
parch	891.0	-	-	0.38159 371492 70	0.80605 722112 99	0.0	6.0
fare	891.0	-	-	32.2042 079685 74	49.6934 285971 80	0.0	512.329 2
embarked	889.0	3	S	-	-	-	-
class	891.0	3	Third	-	-	-	-
who	891.0	3	man	-	-	-	-

Feature	count	unique	top	mean	std	min	max
adult_male	891.0	2	True	-	-	-	-
deck	203.0	7	C	-	-	-	-
embark_town	889.0	3	Southampton	-	-	-	-
alive	891.0	2	no	-	-	-	-
alone	891.0	2	True	-	-	-	-

3. Processing & Transformation Logs

Cleaned Dataset Shape: 784 rows x 17 columns

Outlier Handling: IQR method used to cap numerical outliers.

New Features Created: family_size, is_alone

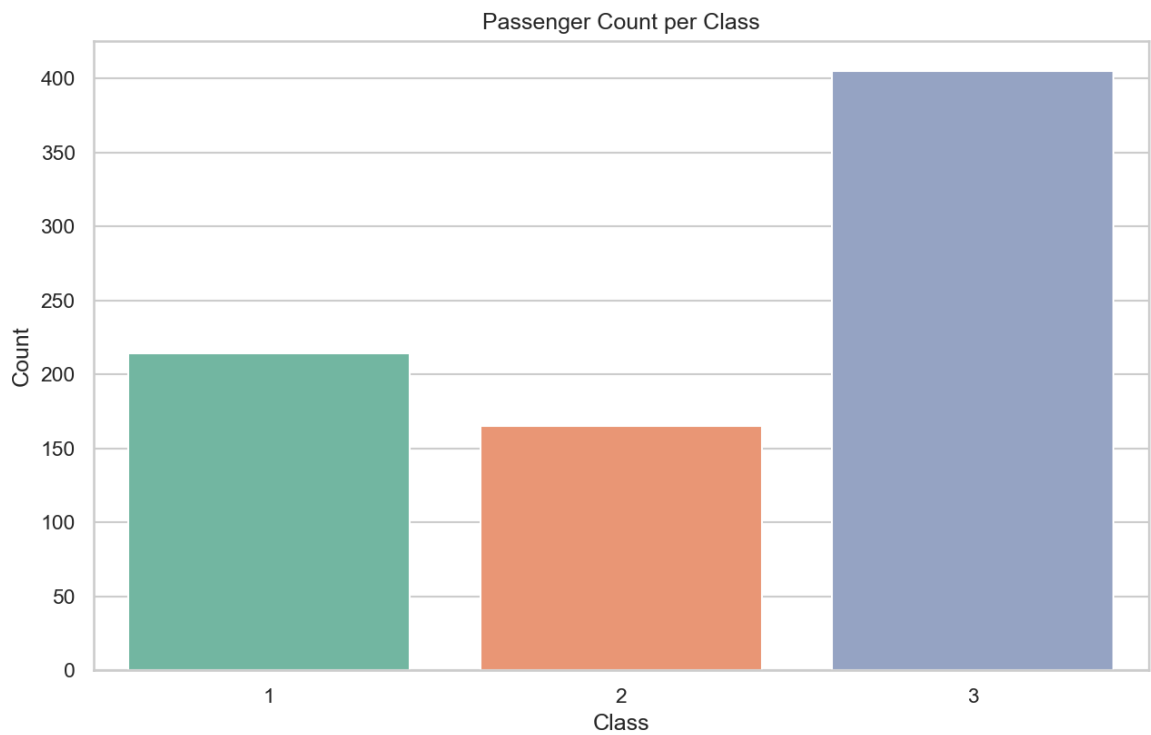
Encoded Categorical Columns: sex, alive

Final Transformed Schema:

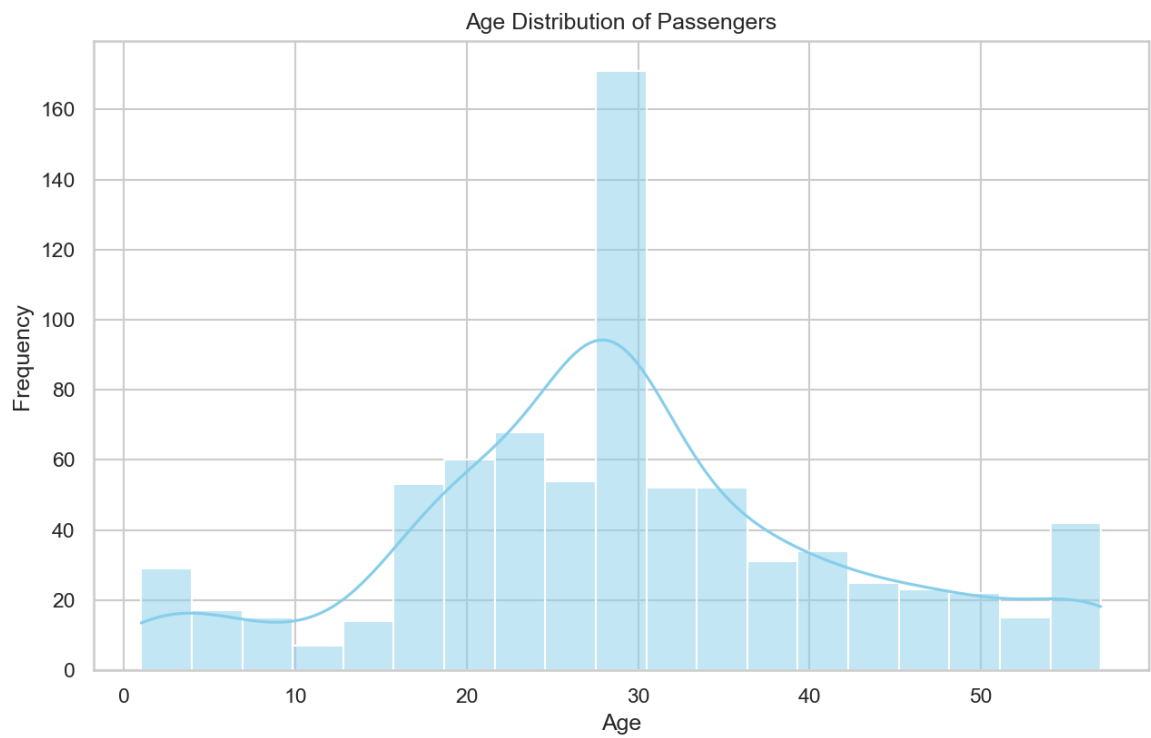
Column	Dtype
survived	int64
pclass	int64
sex	int64
age	float64
sibsp	float64
parch	float64
fare	float64
embarked	object
class	category
who	object
adult_male	bool
deck	category
embark_town	object
alive	int64
alone	bool
family_size	float64
is_alone	int64

4. Data Visualizations

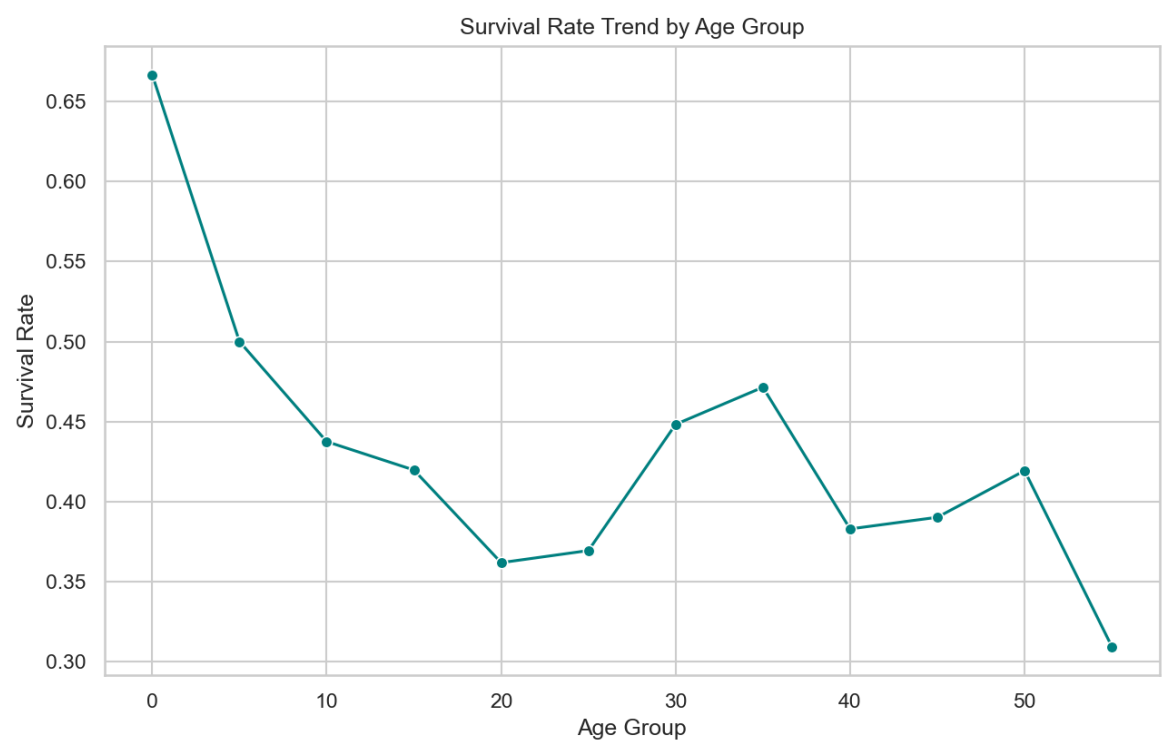
Bar Passenger Class



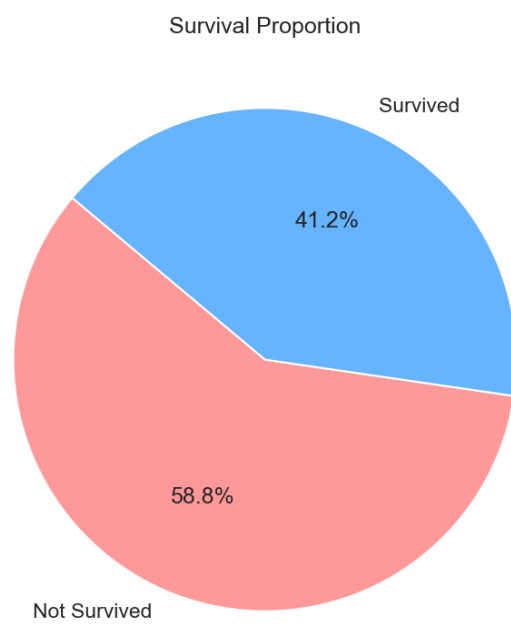
Histogram Age



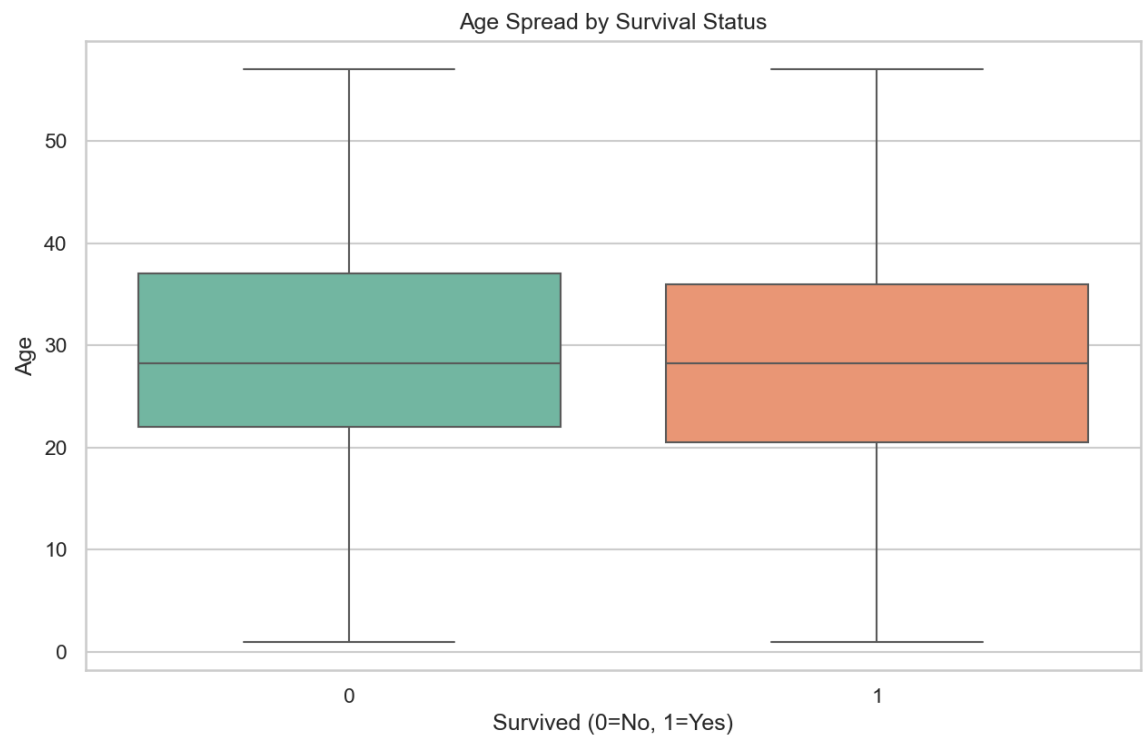
Line Survival Trend



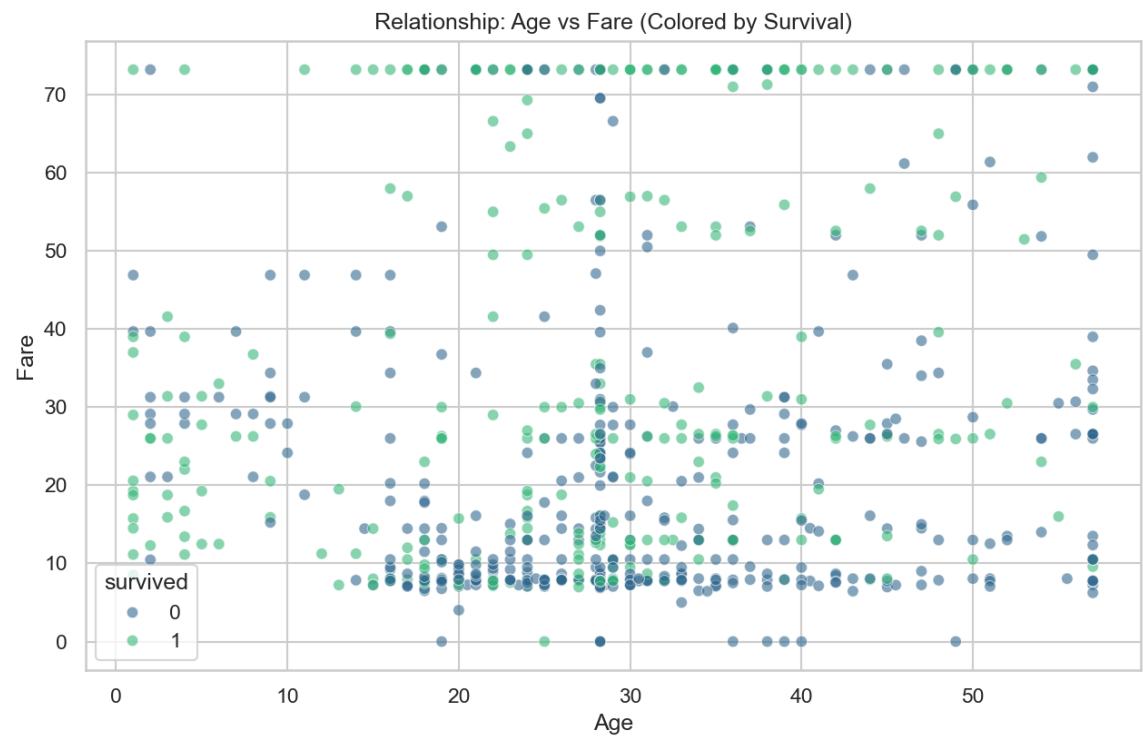
Pie Survival



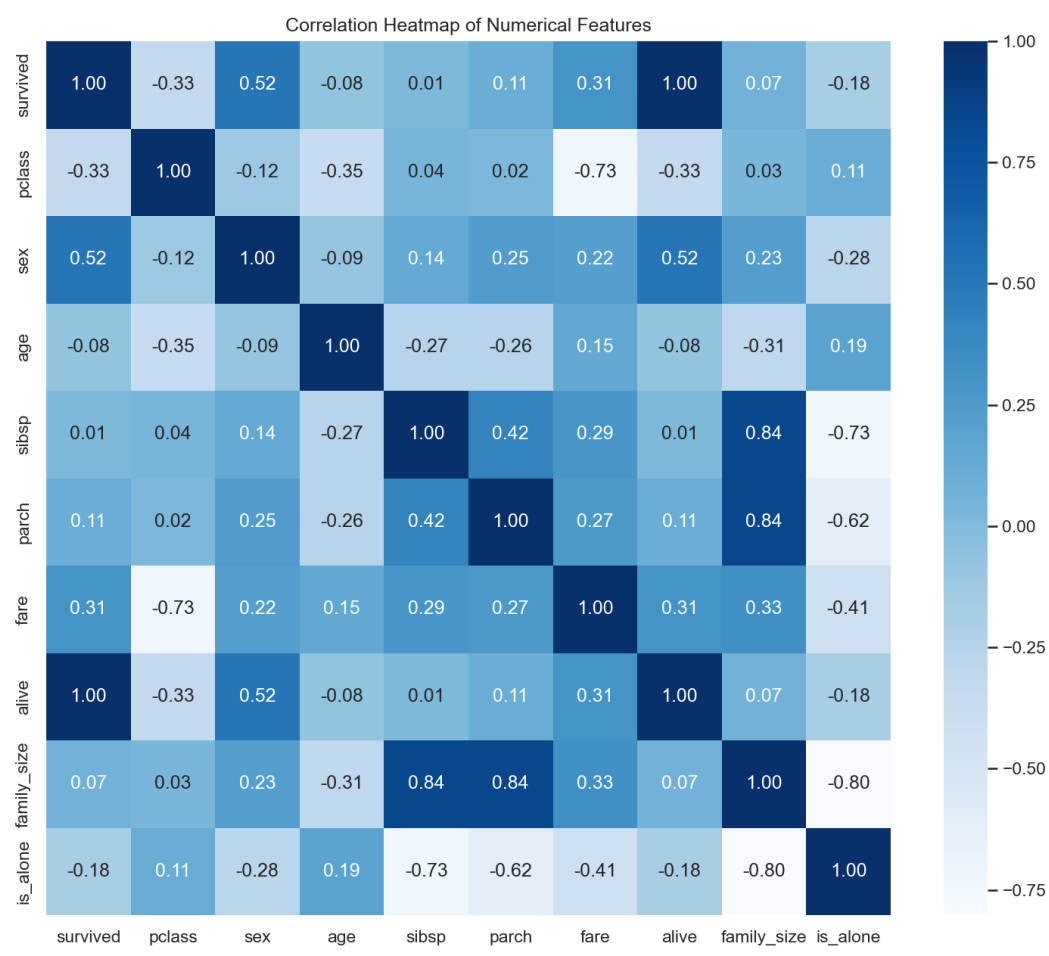
Boxplot Age Survival



Scatter Age Fare



Heatmap Correlation



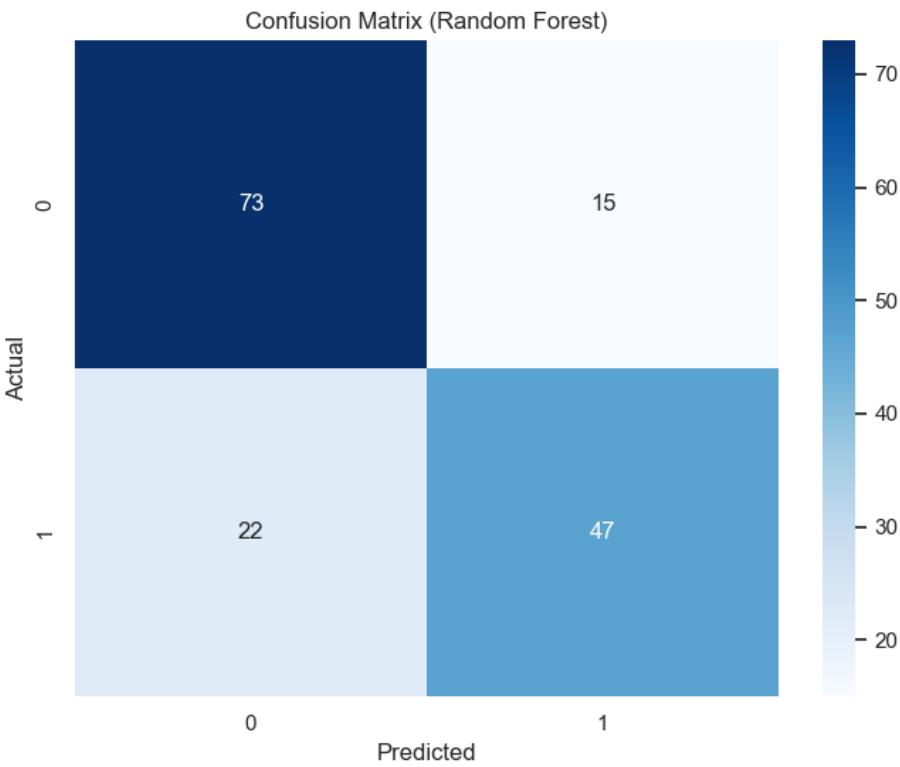
5. Predictive Modeling Results

Models trained to predict the target variable using the cleaned dataset.

Model Performance Comparison:

Model	Metric	Value
Linear Regression	MSE	0.1557
Linear Regression	R2	0.3681
Decision Tree	Accuracy	0.7325
Random Forest	Accuracy	0.7643
Random Forest	AUC	0.8115

Confusion Matrix



Roc Curve

